From Paper to Proof

Getting Lean running · BGSMath, UPC, November 2026

You don't need a laptop in class: the sessions are demonstrations, and you redo them at home. For the first two sessions a web browser is all you need. For sessions 3 and 4 it's worth having Lean on your own laptop, and this sheet takes you through it. If anything here doesn't work, don't fight it: use the browser, and ask me at the next session.

1 In the browser (sessions 1 and 2)

1. Go to `live.lean-lang.org`. There's nothing to install.
2. Open a course file, say `Exercises1.lean`, in the `BGSMath` folder of `github.com/mkaratarakis/from-paper-to-proof`. Click *Raw*, then copy all of it.
3. Back in the browser, delete whatever is in the editor and paste the file in.
4. The first line, `import Mathlib`, takes a minute or two to load the first time. That's normal.
5. The panel on the right is the *infoview*. Click on any line and it shows you what Lean knows there.

The page doesn't save your work, so copy it back into a file before you close it.

2 On your laptop (sessions 3 and 4)

Set aside about half an hour, a good internet connection, and a few gigabytes of disk space. These are the steps from `lean-lang.org/install`, which also has screenshots.

1. Install **VS Code** from `code.visualstudio.com`. It's a free text editor; Lean lives inside it.
2. In VS Code, open the Extensions panel (Ctrl+Shift+X, or Cmd+Shift+X on a Mac), search for `lean4` and install **Lean 4**. A setup guide opens; follow it, and it installs Lean for you.
3. Create a project with Mathlib: click the ∀ symbol at the top right of VS Code, then *New Project…* and *Create Project Using Mathlib…*. Pick a folder and a name, say `bgsmath-lean`. The first time, Mathlib is downloaded, which takes a while.
4. Get the course files from the same address as above: *Code* → *Download ZIP*. Copy the `.lean` files from its `BGSMath` folder into your project folder.
5. **Try it.** Open `Test.lean` from your project folder and click on the line `#eval 2 + 2`. If the infoview shows `4` and nothing is underlined in red, you're done.

One thing that can go wrong: Lean starts "building" thousands of files. Stop it: close VS Code, open a terminal in the project folder, run `lake exe cache get`, and open VS Code again.

3 An AI agent (session 3, optional)

- In session 3 I'll use a coding agent, Claude Code, to write Lean proofs. Other agents work the same way.
- Install instructions: `code.claude.com/docs`. You need an account, and most plans are paid, so this is optional.
- To let the agent see what you see in the infoview, add **lean-lsp-mcp**, following `github.com/oOoooOo/lean-lsp-mcp`.
- No agent? Every exercise can be done by hand.

4 If you're stuck

- Use the browser for now; it's enough for sessions 1 and 2.
- Ask at the start of the next session, or on the Lean Zulip chat, in the stream *#new members*: `leanprover.zulipchat.com`. Questions of every level are welcome there, really.